

SHUN DONG

(513) 835-1178
shun.dong@ku.edu

University of Kansas, School of Business
Lawrence, KS

EDUCATION

University of Kansas	Lawrence, KS
Ph.D. Candidate in Analytics, Information and Operations	GPA: 3.84 Expected May 2027
University of Cincinnati	Cincinnati, OH
M.S. in Statistics	GPA: 3.72/4.0
Missouri State University	Springfield, MO
M.S. in Mathematics, Statistics focus	GPA: 3.82/4.0
M.B.A.	GPA: 3.91/4.0
B.S. in Mathematics, Actuarial Science and Statistics focus	GPA: 3.48/4.0

RESEARCH INTERESTS

Statistical Learning; High-Dimensional Data Analysis; Nonparametric and Semiparametric Statistics; Statistical Computing; Machine Learning Applications in Business, Finance, and Risk Analytics.

JOB MARKET PAPER

Accounting for Industry Heterogeneity in Predicting Corporate Default Probability: A Semiparametric Learning Approach [JOB MARKET PAPER]

with Shaobo Li, Ben Sherwood, and Xiaorui Zhu

Revise and Resubmit, European Journal of Operational Research

Abstract: Predicting corporate default is essential for risk management, credit allocation, and investment decision making, yet most bankruptcy models overlook heterogeneity in how financial distress manifests across industries. We propose the single-index hazard model with multiple links (SIML), a semiparametric learning method that combines a common financial risk index with industry-specific flexible link functions. This structure captures nonlinear industry effects and their interactions with firm-level financial characteristics while retaining an interpretable low-dimensional index. We also develop a two-stage simulation-based testing procedure for industry heterogeneity and nonlinearity. Using a comprehensive database of 18,651 U.S. publicly traded firms from 1980 to 2023, we find strong evidence of distinct nonlinear risk patterns across industries. SIML outperforms a broad set of benchmark models in both in-sample and out-of-sample evaluations. A simulated corporate loan market and an asset-pricing analysis of the distress risk anomaly further show the economic value of accounting for industry-specific default risk.

PUBLICATIONS

Relationship between piperacillin concentrations, clinical factors and piperacillin/tazobactam-associated acute kidney injury

Tang Girdwood, S., Hasson, D., Caldwell, J. T., Slagle, C., **Dong, S.**, Fei, L., Tang, P., Vinks, A. A., Kaplan, J., & Goldstein, S. L. *Journal of Antimicrobial Chemotherapy*, 78(2), 478-487 (2023). DOI: 10.1093/jac/dkac416.

Diagnostic accuracy of serum matrix metalloproteinase-7 as a biomarker of biliary atresia in a large North American cohort

Pandurangi, S., Mourya, R., Nalluri, S., Fei, L., **Dong, S.**, Harpavat, S., Guthery, S. L., Molleston, J. P., Rosenthal, P., Sokol, R. J., Wang, K. S., Ng, V., Alonso, E. M., Hsu, E. K., Karpen, S. J., Loomes, K. M., Magee, J. C., Shneider, B. L., Horslen, S. P., Teckman, J. H., & Bezerra, J. A. *Hepatology* (2024). DOI: 10.1097/HEP.0000000000000827.

Maternal metabolic dysfunction-associated steatotic liver disease during pregnancy and its impact on offspring steatosis

Klepper, C. M., Trout, A. T., **Dong, S.**, Fei, L., Stankiewicz, T. E., Wayland, J., Moreno-Fernandez, M. E., Divanovic, S., Xanthakos, S. A., DeFranco, E., Woo, J. G., & Mouzaki, M. *Clinical Nutrition ESPEN*, 70, 327-335 (2025). DOI: 10.1016/j.clnesp.2025.10.009.

WORKING PAPERS

Cost Restricted Feature Selection with Information Criterion

with Ben Sherwood and Shaobo Li

Preparing for submission to Management Science

Abstract: We study feature selection for data acquisition when candidate variables are costly and must be selected under an acquisition budget. Existing cost-restricted methods typically focus on predictive loss subject to the budget, but they do not explicitly account for the increase in model complexity from acquiring additional variables. We propose a cost-restricted information criterion approach, CRIC, that preserves the acquisition budget as a feasibility constraint while adding an AIC- or BIC-type complexity charge for each selected feature. This allows the method to reject feasible variables whose incremental contribution is too small to justify their inclusion, rather than automatically exhausting the available budget. We develop computational procedures for Gaussian and binary-response settings and evaluate the approach through extensive simulations and empirical applications. The results show that CRIC can substantially reduce unnecessary feature acquisition and select smaller models while preserving competitive out-of-sample predictive performance, with particularly clear gains when the acquisition budget is relatively loose.

Objective-Aligned Sparse Learning for Neyman-Pearson Classification

with Shaobo Li, Ben Sherwood, and Xiaorui Zhu

Preparing for submission to INFORMS Journal on Computing

Abstract: This paper develops an objective-aligned framework for asymmetric classification in which score estimation and variable selection are learned specifically to minimize false negatives subject to a prespecified false positive rate constraint. Standard Neyman-Pearson procedures typically estimate a scoring rule under a symmetric objective and then use threshold calibration to impose asymmetric error control. Our method instead incorporates the deployment objective directly into regularized score learning, while retaining a separate calibration step for final false positive rate control. This allows the selected variables and score direction to reflect the operating point that matters for the decision problem. Simulation studies consistently show lower false negative rates than standard and penalized Neyman-Pearson classifiers, especially when the variables that matter under the asymmetric objective differ from those selected under a symmetric loss. In a bankruptcy application using 18,651 U.S. publicly traded firms from 1980 to 2023, the method reduces the false negative rate from 48.9% to 35.1% at a 5% false positive rate target, a 28% reduction in missed bankruptcies, and raises recall from 51.1% to 64.9%.

Dynamic Instrumented Principal Components for Time-Varying Risk Exposures

with Shaobo Li, and Ben Sherwood

Target journal: Journal of Empirical Finance

Abstract: This paper develops a dynamic extension of instrumented principal components analysis (IPCA) that allows the mapping from firm characteristics to latent factor loadings to evolve over time. Standard IPCA permits characteristics to vary across firms and time but keeps the characteristic-to-loading map fixed. Our model allows this relationship itself to change, capturing persistent shifts in how firm characteristics translate into systematic risk exposures while preserving the characteristic-based factor structure and interpretability of IPCA. Empirically, the dynamic specification substantially improves in-sample fit across latent factor dimensions. With appropriate control of dynamic variation, it also produces sizable out-of-sample gains in return and managed-portfolio prediction relative to static IPCA. The results indicate that allowing the characteristic-risk relationship to evolve improves the description of realized systematic variation and the prediction of expected returns.

Robust Outcome Optimization via Multi-Model Inverse Classification

with Michael Lash, Qihang Lin, and Nick Street

Target journal: Journal of Machine Learning Research

Abstract: This paper develops a multi-model inverse classification framework for robust outcome optimization. Existing inverse classification methods generate intervention recommendations with respect to a single predictive model, even though practitioners often face a collection of plausible models that can disagree about the effect of an intervention. We instead optimize recommendations over the distribution of predictions generated by multiple models and allow the decision maker to choose the desired level of risk. The framework accounts for intervention costs, feasibility constraints, and indirect changes in non-actionable features, while the optimization formulation targets recommendations that remain effective across model variation rather than being tailored to a single fitted model. In an application to cardiovascular risk reduction, the method produces practical recommendations that remain effective across alternative predictive specifications, illustrating how multi-model inverse classification can make prescriptive analytics more robust to variation in the underlying prediction model.

TEACHING EXPERIENCE

University of Kansas, School of Business, Lawrence, KS

2024 to Present

Instructor, Database Management Systems (BSAN 326 / IST 326)

- Independently taught undergraduate database management sections in Fall 2024 and Fall 2025; scheduled to teach again

Shun Dong - 2

in Fall 2026.

- Fall 2024 BSAN 326: instructor evaluation mean 4.29/5; 4.43 for grading clarity, classroom respect, and communication/responsiveness.
- Fall 2025 BSAN 326: instructor evaluation mean 4.10/5; 4.59 for classroom respect and 4.29 for communication/responsiveness.
- Fall 2025 IST 326: instructor evaluation mean 4.36/5; 4.50 for facilitating comprehension and 4.56 for classroom respect and communication/responsiveness.

University of Kansas, School of Business, Lawrence, KS

2023 to Present

Teaching Assistant, Advanced Machine Learning (BSAN 745)

Teaching Assistant, Data Mining and Predictive Analytics (BSAN 450)

University of Cincinnati, Department of Mathematics, Cincinnati, OH

2018 to 2020

Instructor, Elementary Statistics I (STAT 1034) and Elementary Statistics II (STAT 1035)

Teaching Assistant / Recitation Instructor, Calculus I (MATH 1061) and Calculus II (MATH 1062)

Missouri State University, Department of Mathematics, Springfield, MO

2015 to 2017

Teaching Assistant / Lab Instructor, MTH 101 Intermediate Algebra and Calculus

PRESENTATIONS

“Corporate Probability of Default: A Single-Index Model with Multiple-Link Approach”

2026 POMS Annual Conference; Big XII + MIS Research Symposium

2024 Joint Statistical Meetings (JSM)

2023 CMStatistics; INFORMS Annual Meeting

“Robust Outcome Optimization via Multi-Model Inverse Classification”

2025 INFORMS Annual Meeting

SERVICE

Faculty Assembly Representative, KU School of Business

2026–Present

Represent the interests of Ph.D. students in faculty assembly meetings and participate in school-level governance, discussions, and voting.

Ad Hoc Reviewer

2024–Present

INFORMS Workshop on Data Science (2026); INFORMS Data Mining Best Paper Competition (2026); Independent Commodity Intelligence Services (ICIS) (2025); International Workshop on Health Intelligence (2024)

HONORS AND AWARDS

School of Business Best Paper Award, University of Kansas 2026

Max E. Fessler Dissertation Support Award, University of Kansas 2026

John O. Tollefson Best Paper Award, University of Kansas School of Business 2026

Graduate Teaching Assistantship, University of Kansas 2023–Present

Graduate Research Assistantship, University of Cincinnati 2020–2022

Graduate Teaching Assistantship, University of Cincinnati 2018–2020

Graduate Teaching Assistantship, Missouri State University 2015–2017

1st Place, CNAS Thesis Competition Research Award, Missouri State University 2014

CSSA Outstanding Contribution Award 2014

RESEARCH AND PROFESSIONAL EXPERIENCE

Cincinnati Children's Hospital Medical Center, Cincinnati, OH

2020 to 2022

Graduate Research Intern. Collaborated with biostatisticians and physicians on statistical and bioinformatics research, applying advanced statistical methods to investigate disease and chronic-condition risk factors.

China Continent Insurance LLC, Linyi City, Shandong Province, China

Summer 2013 and Summer 2014

Manager Assistant. Assessed vehicle condition for depreciation. Followed up with the insured individuals regarding premium and deductibles payments.

ADDITIONAL INFORMATION

Programming and Computing: R (Rcpp, RcppArmadillo), Python, C, C++, SQL, SAS, MATLAB, LaTeX, Microsoft Office;

Other Experience and Certification: Insurance Agent Qualification Certificate, 2013

REFERENCES

Shaobo Li

Associate Professor, Analytics, Information, Operations
University of Kansas School of Business | shaobo.li@ku.edu

Ben Sherwood

Associate Professor; Jack and Shirley Howard Mid-Career
Professor
University of Kansas School of Business | ben.sherwood@ku.edu

Debabrata "Deb" Dey

Davis Area Director, Analytics, Information, Operations; Ronald
G. Harper Professor of Artificial Intelligence and Information
Systems
University of Kansas School of Business | deb.dey@ku.edu

Lin Fei, Ph.D.

Professor, UC Department of Pediatrics
Cincinnati Children's Hospital Medical Center |
lin.fei@cchmc.org